| Question | Answer | Marks | Partial Marks |
|---|---|---|---|
| 1 | 0.0001 | **1** | Allow $1 \times 10^{-4}$ or 1/10000 |
| 2 | 57.40 cao | **1** | |
| 3 | 275 | **1** | |
| 4 | 38 | **2** | **M1** for 180 – 61 – 81<br>or for angle [$BAC$ =] 180 – 2 × 61 or better<br>soi by 58<br>or for angle [$BAD$ =] 180 – 61 – (180 – 81) or better<br>soi by 20 |
| 5 | 1700 | **1** | |
| 6 | $6\frac{2}{3}$ oe | **2** | **M1** for figs 666 or 667 |
| 7(a) | $\begin{pmatrix} 27 \\ -12 \end{pmatrix}$ | **2** | **B1** for $\begin{pmatrix} 27 \\ k \end{pmatrix}$ or for $\begin{pmatrix} k \\ -12 \end{pmatrix}$ |
| 7(b) | 13 | **2** | **M1** for $(-12)^2 + 5^2$ oe soi by 169 |
| 8 | $9 + 4.5\pi$ oe | **3** | **M2** for $\frac{9\pi}{2}$ oe<br>or **M1** for $9\pi$ oe |
| 9(a) | –3 | **1** | |
| 9(b) | $22 - 5n$ oe final answer | **2** | **B1** for $k - 5n$ or for $22 - kn$ $k \neq 0$ as final answer or for $22 - 5n$ oe seen then further incorrect working |
| 10 | $6\frac{1}{6}$ cao | **3** | **B2** for $\frac{37}{6}$<br>or **M2** for $\frac{16}{6} + \frac{21}{6}$ or for $5\frac{4}{6} + \frac{3}{6}$<br>or **M1** for $\frac{8}{3} + \frac{7}{2}$ or for $2 + 3 + \frac{2}{3} + \frac{1}{2}$ |
| 11 | 16 | **2** | **B1** for 4 seen |
| 12(a) | $1.42 \times 10^{-11}$ | **2** | **B1** for figs 142 or $10^{-12}$ seen |
| 12(b) | $5.72 \times 10^{7}$ | **2** | **M1** for figs 572 |

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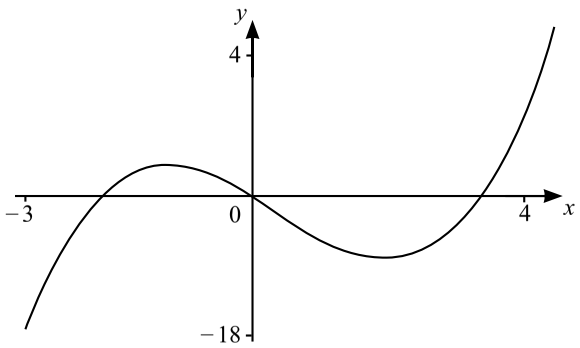
| Question | Answer | Marks | Partial Marks |
|---|---|---|---|
| 22 | $\frac{16}{\sqrt{3}}$ or $\frac{16\sqrt{3}}{3}$ | 4 | **B1** for $\cos 30 = \frac{\sqrt{3}}{2}$<br>**M2** for $\frac{8}{\cos 30}$<br>or **M1** for $\frac{8}{x} = \cos 30$ |
| 23 | $\frac{25 - x}{(x - 4)(x + 3)}$ oe final answer | 3 | **M1** for denominator $(x – 4)(x + 3)$ or $x^2 – x – 12$<br>**B1** for $3(x + 3) – 4(x – 4)$ seen |
| 24(a) | $(x – 2)^2 + 3$ | 2 | **M1** for $(x – 2)^2$ seen |
| 24(b) | (2, 3) | 1 | **FT** *their* **(a)** |
| 25 | $\frac{15}{2}$ oe | 3 | **M2** for $\frac{9}{x} = \frac{6}{5}$ oe or $x^2 = \frac{25}{36} \times 9^2$ oe or better<br>or **M1** for $\sqrt{\frac{36}{25}}$ or $\sqrt{\frac{25}{36}}$ oe or better<br>or for $\left(\frac{9}{x}\right)^2 = \frac{36}{25}$ or $\left(\frac{x}{9}\right)^2 = \frac{25}{36}$ oe or better |
| 26(a) | Correct sketch and –2, 0 and 3 indicated on $x$-axis with graph going beyond intersections with $x$-axis<br>  <br> | 3 | **B2** for correct sketch without labels<br>or **B1** for any positive cubic with two turning points |
| 26(b) | $x^3 – x^2 – 6x$ | 3 | **B2** for $x(x^2 + 2x – 3x – 6)$ or $(x^2 + 2x)(x – 3)$ or $(x + 2)(x^2 – 3x)$<br>or **B1** for $x^2 + 2x – 3x – 6$ or $x^2 + 2x$ or $x^2 – 3x$ |

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| Question | Answer | Marks | Partial Marks |
|---|---|---|---|
| 26(c) | (0, –1) | **5** | **B2** for $3x^2 - 2x - 6$<br>or **B1** for $3x^2$ or $2x$ or – 6<br><br>**B2** for $y = -5x - 1$<br>or<br>**M1** for substituting 1 into *their* derivative<br>and **M1dep** for $-6 = (\textit{their}\ -5) \times 1 + c$ or use of rise/run to find rise<br><br>**FT** *their* **(b)** for the whole of **(c)** |

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